

Devis Saputra

Learning & Development | AI in Education Research

Medan, Indonesia | Open to remote work and relocation

devis.saputra@gmail.com | +62 812 9321 7351

Portfolio: devissaputra.github.io | LinkedIn: linkedin.com/in/devissaputra

ORCID: [0000-0002-7133-9410](https://orcid.org/0000-0002-7133-9410) | Academic CV

RESEARCH PROFILE

Learning and development practitioner and former lecturer with experience in curriculum design, learning systems, assessment and program leadership. My research interests focus on interpretable AI for learner support, assessment and workplace learning. Recent portfolio studies use Python to examine learner modeling, retrieval, predictive risk and model calibration through baseline comparisons, leakage controls and uncertainty analysis. I aim to connect technical performance with educational validity and the practical conditions of implementation.

EDUCATION

Master of Business Administration (MBA), International Business

University of Applied Sciences Würzburg Schweinfurt (now THWS), Germany | October 2012 to June 2014 | Final grade: 1.8, German scale

Thesis: Leadership on Flat Organizational Structure: Focusing on Organization Using Self Managing Team and Open Allocation Strategy. Comparative case study of Valve and GitHub.

Bachelor of Engineering (BEng)

Institut Teknologi Bandung, Indonesia | August 2004 to March 2010

Thesis: Image Retrieval System Using Geotagged Information. Developed an image retrieval approach combining geotagged EXIF metadata with vector space text retrieval.

SELECTED EMPIRICAL STUDIES | 2026

Reproducible portfolio studies with public code, documented data provenance and manuscript drafts. These are research outputs in development, not peer reviewed publications.

AI in Education devissaputra.github.io/aieducation.html

- Knowledge Tracing Benchmark on ASSISTments 2009: 4,148 learners, 274,331 interactions and learner disjoint evaluation; mean ROC AUC across three GRU seeds was 0.7471 versus 0.6982 for PFA.
- Early Assessment Submission and Later Outcomes in OULAD: 1,820 cases in a day 30 landmark design; weighted Hájek contrast 0.3861 with a bootstrap interval of about 0.3242 to 0.4447, interpreted as observational.
- Wrong Answer Conditioned Retrieval on SciQ: 884 questions and 2,652 wrong answer proxy cases; BM25 MRR 0.9472 for question only retrieval versus 0.9433 with wrong answer expansion.
- Enrollment Time Dropout Risk Model Audit: 4,424 records and 20 enrollment time predictors; holdout ROC AUC 0.8316 and Brier score 0.1447 with explicit subgroup audit limits.

Learning & Development devissaputra.github.io/learningdevelopment.html

- Organizational Communication Resilience Under Key Actor Loss: Enron network with 148 actors and 10,885 events; 30 targeted removals retained 0.8608 of baseline efficiency versus 0.9917 for comparators.
- Employer Training Investment Capacity: UK Employer Skills Survey from 2011 to 2024; real spend per employee fell from £2,410 to £1,700, a 29.46% decline, while participation reach increased.
- Technology Skill Renewal Pressure: O*NET from 2020 to 2026 across 902 occupations; median software skill Jaccard 0.6923 and 13.97% of occupations below 0.50.
- Decision Value of Training Evidence: National Supported Work reanalysis with 185 treated participants and 260 controls; mean RE78 difference +\$1,794.34 and bootstrap 95% interval from \$519 to \$3,118.

AI Engineering devissaputra.github.io/aiengineering.html

- Probability Calibration on UCI Bank Marketing: 45,211 observations across five fixed holdouts; isotonic mean Brier score 0.0693 and ECE 10 of 0.0074 while ROC AUC stayed near 0.906.
- Label Blind Threshold Aware Anomaly Detection on NAB: four chronological streams; a 5% validation alert budget produced future test false positive rates from 0.0300 to 0.2338.
- Explainable AI: Model Dependence and Feature Governance: 48,842 UCI Adult records; random forest ROC AUC 0.9174 with all features and 0.9167 after protected feature exclusion; permutation importance remained model dependent.
- Transformer Forecasting on WDC SILSO: 3,327 monthly observations; Transformer MAE 15.914 at one month, 19.691 at six months and 24.206 at twelve months, showing horizon dependent performance.

System Engineering devissaputra.github.io/research.html#system-engineering

- Empirical Engineering Design Trade Space: 103 UCI concrete slump experiments; the baseline Pareto frontier contained 23 observations, falling to 10 when slump was removed as an objective and to 8 under a 20 cm eligibility threshold.
- Requirements Classification Evidence Triage: evaluated 571 eTour requirement elements; label F1 ranged from 0.653 to 0.945, while a prediction-only queue captured 115 of 534 label errors in the first 100 reviews versus 17.5% expected under random review.
- Forecast Maturity of Technical Performance Margin: analyzed four NASA lithium-ion battery trajectories; mean absolute threshold-cycle error across three crossing cells fell from 102.399 cycles at checkpoint 40 to 11.942 at checkpoint 80.
- Cross-Project Heterogeneity in Quality-Oriented Engineering Change: reanalyzed 2,533 manually classified commits from 54 Apache projects; pooled maintenance shares masked persistent cross-project heterogeneity, with Cramér's V remaining near 0.30 across sample-size restrictions.

ACADEMIC & PROFESSIONAL EXPERIENCE

Independent Learning Consultant

Independent projects / Remote, Indonesia | May 2025 to present

- Develop learning needs analyses, competency maps, onboarding structures, learning paths, job aids and assessment plans for independent learning projects.
- Conduct research synthesis and develop independent Python studies in learner modeling, retrieval evaluation, learning analytics and model evaluation, with explicit attention to reproducibility and evidence limits.

Learning and Development Manager

Suitmedia, Jakarta; remote | February 2023 to August 2024

- Led Rocketiversity, a customized Moodle learning system, from content audit and course structuring through manager approval and rollout with a project team of four people.
- Established 120 SCORM compliant courses and structured onboarding for around 100 employees; reported a 30% reduction in onboarding time in the project case study.
- Converted fragmented departmental materials into reusable learning units; clarified content ownership, trainer responsibilities and content maintenance practices.

Lecturer in Engineering Management

Institut Teknologi Del, Laguboti | May 2016 to June 2023

- Taught engineering economics, entrepreneurship and systems thinking, and coordinated internship and capstone learning through applied problems and project based assessment.
- Supervised more than 50 undergraduate theses and capstone projects; developed internship guidance, assessment structures and practical work documentation.
- Contributed to the Astra Lean Production System teaching laboratory and co designed practicum activities.

Head of Engineering Management Program

Institut Teknologi Del | November 2017 to May 2021; concurrent with lecturer role

- Led the 2019 to 2024 curriculum revision, translating expert consultation, graduate feedback and industry needs into a 144 SKS, eight semester program.
- Aligned learning outcomes, course sequencing and assessment; coordinated peer review, Academic Senate approval and course equivalency rules for student progression.
- Led and mentored eight lecturers, connecting curriculum governance, teaching quality and industry linked learning.

Program Manager, Health Technology Initiative

Summit Institute for Development, Mataram | May to October 2015

- Directed four developers and six field workers across 16 districts for the THRIVE/OpenSRP mobile health workflow initiative.
- Used health worker interviews and field timing measurements to standardize form completion and diagnose XLSForm latency; an interim form splitting solution reduced most form loads to under 10 seconds.

Research and Development Engineer

PT Cipta Srigati Lestari, Jakarta | February to August 2011

- Developed Java based optimization tools for B2B supply chain and operational decision support problems, translating business requirements into computational solutions.

PUBLISHED PAPERS & CONFERENCE PROCEEDINGS

Silalahi, F. T. R., **Saputra, D. W.**, Panjaitan, G., Butar Butar, M., & Manurung, N. M. (2023). **Influencing Student Choice: A Case Study of a Boarding Private University in Indonesia.** Proceedings of ICEBE 2022, 429 to 443. DOI

Saputra, D. W. (2022). Optimalisasi Rute Distribusi Kurir Menggunakan Metode Traveling Salesman Problem (Studi Kasus: JNE Balige). [**Courier Distribution Route Optimization Using the Traveling Salesman Problem Method: A Case Study of JNE Balige.**] G Tech: Jurnal Teknologi Terapan, 6(2), 159 to 165. DOI

Meliala, G. N., & **Saputra, D. W.** (2020). Usulan Tata Letak Gudang Non Semen Menggunakan Metode Class Based Storage di PT Intiaga Sukses Abadi Cabang Medan. [**Proposed Non Cement Warehouse Layout Using the Class Based Storage Method at PT Intiaga Sukses Abadi Medan Branch.**] Jurnal Industri Kreatif, 4(1), 17 to 26. DOI

Siboro, B. A. H., Sinaga, R. H. M., & **Saputra, D. W.** (2019). Rancang Bangun Alat Pengering Andaliman dengan Menggunakan Metode Quality Function Deployment (QFD). [**Design and Development of an Andaliman Dryer Using the Quality Function Deployment (QFD) Method.**] Jurnal Sains dan Teknologi, 19(2), 133 to 140. DOI

Sinaga, R. H. M., & **Saputra, D. W.** (2019). **Performance Investigation of Geothermal Power Plant Based on Exergy and Exergoeconomic Analyses (Case Study: Sarulla Geothermal Power Plant).** IOP Conference Series: Materials Science and Engineering, 546, 072006. DOI

Tampubolon, S. L., **Saputra, D. W.**, & Simanjuntak, M. (2019). Analisis Kepuasan Wisatawan terhadap Atribut Wisata Menggunakan Metode CSI dan IPA pada Wisata Pemandian di Kabupaten Tapanuli Utara. [**Analysis of Tourist Satisfaction with Tourism Attributes Using CSI and IPA Methods at Bathing Attractions in North Tapanuli Regency.**] Jurnal Sains dan Teknologi, 19(2), 141 to 151. DOI

BOOKS

Saputra, D. W., Siboro, B. A. H., & Sinaga, C. (2022). Metodologi Pelaksanaan Kerja Praktik di Bidang Manajemen Rekayasa [Methodology for Conducting Practical Work in Engineering Management].

Saputra, D. W. (2019). Curriculum of Engineering Management Program, Del Institute of Technology, 2019 to 2024. Institutional curriculum document.

ACADEMIC SERVICE & SUPERVISION

- Senate Member, Engineering Management Department, 2016 to 2018.
- Coordinated student internship processes and practical work documentation.
- Supervised more than 50 undergraduate theses and capstone projects.
- Copyright co-holder, Andaliman Dryer Machine

COMPETENCE

Learning Design & Curriculum: needs analysis; learning objectives; curriculum and learning path design; project based assessment; blended and experiential learning.

Learning Systems & Governance: Moodle; SCORM; course structuring; content audit and standardization; subject matter collaboration; content ownership and maintenance.

Research & Evaluation: case studies; descriptive and comparative analysis; causal and observational study design; baseline comparisons; model calibration; error analysis; bootstrap uncertainty; reproducible reporting.

AI & Data: Python; NumPy; pandas; scikit-learn; PyTorch; classification; sequence models; retrieval evaluation; learner modeling; leakage checks; calibration; Git and GitHub for versioned research code.

LANGUAGES

Indonesian and Batak: Native. English: Fluent. German: Working Proficiency.